Missing Data Imputation in ECG Signals:

A Comparative Study of Recurrent Models and Multi-Scale Transformers

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Abstract

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Missing values are a fundamental obstacle in ECG signal analysis, arising from motion artifacts, electrode dropout, and transmission loss during acquisition. Corrupted or incomplete signals directly compromise downstream clinical tasks such as arrhythmia detection and rhythm classification, making high-quality imputation a medically significant problem. ECG signals are particularly challenging to impute due to their sharp QRS complexes, patient-specific morphology, and non-stationary character — properties that defeat simple interpolation assumptions.

This project presents a systematic comparison of imputation methods across three tiers of complexity, evaluated on the MIT-BIH Arrhythmia Database (48 records, 360 Hz). The first tier comprises six statistical baselines including linear interpolation, cubic spline, PCHIP, and Savitzky-Golay smoothing. The second tier introduces two bidirectional RNN models — BiLSTM and BiGRU — trained with an amplitude-weighted loss that emphasizes reconstruction near R-peaks. The third tier is our proposed model: a two-stage SAITS-style transformer with multi-scale patch embeddings (kernel sizes 4, 8, and 16), stochastic depth regularization, a learnable mask token, and curriculum masking that progressively transitions from random dropout to clinically demanding block-target-peak missingness.

Among all baselines, BiGRU achieves the strongest performance (MAE = 0.5073, PRD = 44.3%), while BRITS, despite its bidirectional recurrent design, lags substantially (MAE = 0.7261, PRD = 67.9%), partly due to computational constraints that limited its training data. A consistent finding across all methods is that QRS-region MAE is two to three times higher than flat-region MAE, confirming that R-peak reconstruction remains the central difficulty. Our transformer reduces this gap, achieving the best overall performance (MAE = 0.4824, PRD = 43.45%) and improving upon the best RNN baseline. This improvement is supported by bootstrap 95% confidence intervals and per-patient analysis across eight test subjects. Ablation studies isolate the contributions of multi-scale patching, curriculum scheduling, and stochastic depth, showing that each component contributes meaningfully to the final result.

The study concludes that careful architectural design — particularly multi-scale temporal representation and curriculum-guided training — yields measurable and statistically robust improvements over recurrent baselines on this clinically motivated imputation task.

I Introduction

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I.1 Context and Motivation

Electrocardiogram (ECG) signals are among the most diagnostically important time-series in clinical medicine [14], [15]. Continuous ECG monitoring underpins the detection of arrhythmias, ischemic events, and conduction abnormalities, making signal integrity a direct prerequisite for reliable diagnosis. In practice, however, recorded ECG signals are frequently incomplete. Motion artifacts corrupt waveforms during ambulatory monitoring, electrode detachment introduces sudden dropout, and wireless transmission losses produce contiguous missing segments. Preprocessing pipelines add a further layer of removal by rejecting windows whose kurtosis or signal-to-noise ratio falls below acceptable thresholds. The result is that even carefully curated datasets contain gaps that must be filled before any downstream analysis can proceed reliably [16].

Imputation — estimating plausible values at missing positions while remaining consistent with the observed signal — is the standard response to this problem. Classical approaches such as linear interpolation, spline fitting, and last-observation-carried-forward are computationally inexpensive and easy to apply [19], [18], but they impose overly smooth assumptions on a signal whose most clinically relevant feature, the QRS complex, is a sharp, high-amplitude transient lasting only 80–120 ms. These methods reconstruct flat and baseline regions acceptably but systematically fail at the very morphological events that carry diagnostic information.

Deep learning offers a principled alternative. Recurrent architectures can learn temporal dependen-

cies from data rather than relying on fixed interpolation rules [1], [2], and attention-based transformers can integrate context from both sides of a missing segment simultaneously [4], [3]. Yet added model complexity is only justified if it translates into measurable, reproducible improvement over strong baselines — particularly on the QRS regions that matter most clinically.

This project investigates that question directly, using the MIT-BIH Arrhythmia Database [11] as a controlled experimental setting. We compare statistical baselines, bidirectional RNN models, and a proposed transformer architecture designed around the specific challenges of ECG imputation: sharp morphological events, patient-specific waveform structure, and clinically motivated masking patterns that simulate realistic signal loss.

I.2 Problem Challenges

ECG imputation is harder than general time-series imputation for three compounding reasons. The first is morphological sharpness. The QRS complex — the dominant feature of the cardiac cycle — rises and falls within roughly 100 ms, producing amplitude excursions that can reach five to ten times the baseline level [13]. Any imputation method that smooths across a missing QRS window does not merely introduce numerical error; it erases the very event that clinicians use to assess heart rhythm. Linear and spline-based methods are structurally incapable of reconstructing this kind of sharp transient from neighboring observations alone [19], [18].

The second challenge is patient-specific morphology. ECG waveform shape varies substantially across individuals due to differences in cardiac anatomy, electrode placement, and underlying pathology [14]. A method that learns average waveform structure from population data may reconstruct a plausible-looking signal that is nonetheless wrong for a specific patient. This is particularly relevant for the MIT-BIH Arrhythmia Database [11], which includes patients with complex arrhythmias whose beat morphology deviates significantly from normal sinus rhythm.

The third challenge is non-stationarity. ECG signals are not stationary processes — heart rate, amplitude, and baseline wander all change over time within a single recording. A model trained on one segment of a patient's signal cannot assume that the statistical properties of that segment hold elsewhere. This makes it difficult to rely on global statistics and pushes toward modern architectures, such as self-attention mechanisms, that can adapt their reconstruction based dynamically on local observed context [4], [3].

Together these three properties explain why the QRS-region MAE is consistently two to three times higher than flat-region MAE across all methods in our experiments, and why closing that gap is the central modeling challenge this project addresses.

I.3 Objectives and Scope

The primary objective of this project is to determine whether a transformer-based architecture, specifically designed around ECG signal properties, can achieve meaningfully better imputation than both classical statistical methods and bidirectional RNN baselines under a shared and reproducible evaluation protocol [1], [3]. This comparison spans three tiers: six statistical baselines, two bidirectional RNN models trained with amplitude-weighted loss, and a proposed two-stage SAITS-style transformer with multi-scale patch embeddings, stochastic depth regularization, and curriculum masking [4], [21], [22].

A secondary objective is to evaluate honestly rather than favorably. Aggregate metrics alone can obscure important failure modes, so evaluation is decomposed into QRS-region and flat-region error separately, supplemented by bootstrap 95% confidence intervals and per-patient performance analysis across the test set [24]. Ablation studies further isolate the contribution of individual architectural choices so that any performance gain can be attributed to specific design decisions rather than generic model size.

The scope of the project is intentionally constrained. All experiments use a single lead (MLII) from the MIT-BIH Arrhythmia Database [11] and a fixed window size of 512 samples. RNN baselines were trained on a 40,000-window subsample due to session memory constraints, and BRITS was trained under a separate pipeline due to PyPOTS [28] CUDA compatibility issues with the P100 GPU environment. These limitations mean that results should be interpreted as evidence within

this specific experimental setting rather than as universal conclusions about ECG imputation. The findings are nonetheless meaningful within their scope: the dataset is a standard benchmark, the evaluation protocol is consistent across methods, and the architectural choices are motivated by the domain rather than by arbitrary complexity.

II Background

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II.1 ECG Signal Structure and Why It Matters for Imputation

The electrocardiogram (ECG) records cardiac electrical activity as a continuous waveform of repeating features: the P-wave, QRS complex, and T-wave [14]. Of these, the QRS complex is the most diagnostically significant and difficult to impute. At 360 Hz, it spans roughly 29 to 43 samples—a sharp, high-amplitude transient used to identify arrhythmias, bundle branch blocks, and myocardial infarction [13]. Despite its brief duration, it carries the majority of clinically actionable information in the signal.

This creates a fundamental asymmetry for imputation. The baseline segments surrounding each beat — the PR interval, ST segment, and inter-beat baseline — are smooth and slowly varying, making them recoverable by interpolation from neighboring observations [16]. The QRS complex is not. It rises steeply, peaks, and returns within 30 to 40 samples, producing a gradient that no smooth curve fitted between boundary values can replicate. A method that reconstructs flat regions accurately while failing on QRS peaks may report a low aggregate error while being clinically useless. This is why our evaluation explicitly separates QRS-region MAE from flat-region MAE, and why the gap between these two numbers — consistently a factor of two to three across every method we test — is the central diagnostic of model quality in this project [24].

Two further properties compound the difficulty. First, ECG morphology is patient-specific: duration, amplitude, and axis vary across individuals due to cardiac anatomy and pathology. The MIT-BIH Arrhythmia Database [11], forming the basis of this study, contains complex arrhythmias whose beat morphology deviates substantially from population averages, a failure mode that aggregate metrics can obscure. Second, the ECG is non-stationary, with heart rate and baseline wander drifting over time. In our pipeline, sub-cardiac-rate drift and powerline interference are addressed by a fourth-order Butterworth bandpass filter (0.5–40 Hz) and a 50 Hz notch filter, preserving the full diagnostically relevant frequency range [30].

The MIT-BIH database consists of 48 half-hour two-lead recordings from 47 patients sampled at 360 Hz [32]. With four paced-beat records excluded, the preprocessing yielded 44 records and 171,786 quality-filtered 512-sample windows. This dataset provides a clinically realistic testbed whose prevalence of abnormal morphologies makes it a meaningful benchmark for evaluating whether a proposed architecture genuinely handles difficult cases rather than routine beats.

II.2 Imputation Methods: From Interpolation to Attention

Imputation methods for time-series data can be arranged along a spectrum of complexity, from deterministic analytical rules to fully learned neural architectures [16]. This project evaluates methods at three points on that spectrum, and understanding the conceptual basis of each tier is necessary for interpreting why one outperforms another in particular signal regions.

At the simplest end, statistical baselines make no use of training data and apply a fixed rule to each missing segment independently. Mean imputation replaces missing values with the mean of the observed positions in the same window, ignoring all temporal ordering. Last Observation Carried Forward (LOCF) propagates the most recently observed sample into subsequent missing positions, producing flat segments that are only appropriate when the signal changes very slowly. Linear interpolation connects the last observed value before a gap to the first observed value after it with a straight line, and cubic spline interpolation does the same with a smoother piecewise polynomial that enforces continuous first and second derivatives at the boundaries [19]. PCHIP interpolation is a monotonicity-preserving variant that avoids the overshoot artifacts cubic splines can introduce

near sharp transitions [18]. Savitzky-Golay filtering applies a least-squares polynomial fit within a sliding window, acting as a smoothing operator rather than a true interpolator [17]. All six methods are included in our evaluation because they establish a clear performance floor: any learned model that does not substantially outperform them on QRS regions has not justified its computational cost.

Bidirectional recurrent neural networks represent the next tier. The key limitation of statistical baselines in the QRS region is that they can only use the signal values immediately adjacent to the gap — and those boundary values do not contain enough information to reconstruct a sharp transient of unknown morphology. A recurrent model trained on many ECG windows can instead learn the statistical distribution of QRS shapes from the training data and use that learned prior, combined with the observed context in the current window, to produce a more informed reconstruction [2]. Bidirectional processing is important here because it allows the model to use both the signal before and after a missing segment simultaneously [9], [10]. In our implementation both BiLSTM and BiGRU use hidden size 256, two layers, and dropout 0.3, and are trained with an amplitude-weighted loss that applies a penalty factor of 3.0 to errors at positions where signal amplitude exceeds 1.5 standard deviations, explicitly directing the model's learning capacity toward QRS reconstruction. Both models use early stopping with patience 15 on validation loss and are trained on a 40,000-window subsample of the training set due to session memory constraints.

Transformer architectures take a qualitatively different approach. Rather than propagating information through a sequential hidden state, self-attention computes direct pairwise relationships between every pair of positions in the input sequence simultaneously [4]. For imputation this has a concrete advantage: a missing QRS complex can be directly related to structurally similar beats elsewhere in the same window, without the signal having to travel through a chain of intermediate recurrent states that may attenuate or distort it [5]. Self-attention also has no inherent directionality, integrating past and future context symmetrically without requiring separate forward and backward passes. These properties make attention-based architectures a natural candidate for reconstructing sharp, localized waveform features in a signal where the relevant context may be distributed non-locally across the window — which is precisely the structure of an ECG with multiple beats per 512-sample window [3].

II.3 BRITS and the SAITS Architecture

BRITS — Bidirectional Recurrent Imputation for Time Series — differs from a standard bidirectional RNN applied naively to imputation in one important way: rather than treating missing values as a preprocessing problem to be solved before the model runs, BRITS integrates estimation of missing values directly into the recurrent computation [1]. At each time step where an observation is absent, the model substitutes its current hidden state estimate in place of the missing input and continues the recurrent pass, so that subsequent hidden states are conditioned on a plausible reconstruction rather than on a zero or mean fill. This is done in both the forward and backward temporal directions, and the two directional estimates are combined to produce the final reconstruction at each missing position. BRITS additionally incorporates learned time-delta features that encode the elapsed time since the last valid observation, allowing the model to modulate its uncertainty based on gap length — a longer gap warrants less confident reconstruction than a short one [8]. In our experiments BRITS is implemented via the PyPOTS library [28] with hidden size 64, batch size 128, Adam optimizer with learning rate 0.001 [26], and trained for 20 epochs on 25,000 training windows. A practical limitation was that PyPOTS encountered CUDA compatibility issues with the P100 GPU environment used for training, which constrained both the training set size and the number of epochs relative to what would otherwise have been used. These constraints are acknowledged when interpreting BRITS results in Section IV.

The SAITS architecture — Self-Attention-based Imputation for Time Series — replaces the recurrent backbone with a two-stage transformer pipeline designed specifically for the imputation setting [3]. In Stage 1, the input signal is first embedded using a learned projection, with observed positions represented by their embedded values and missing positions replaced by a learned mask token — a trainable vector parameter that the model

can use to signal that no observed information is available at that position [**?**]. Sinusoidal positional encodings are added to preserve temporal ordering, and the sequence is then passed through a stack of self-attention blocks [4]. Each block applies multi-head self-attention followed by a position-wise feed-forward network, with pre-layer normalization and residual connections throughout. Stage 1 produces a first estimate of the complete signal, including reconstructed values at all missing positions.

In Stage 2, this first estimate is fused back with the original observed signal: at observed positions the true value is retained, and at missing positions the Stage 1 estimate is substituted. This refined sequence — which is now complete everywhere — is re-embedded and passed through a second identical transformer stack, which can now attend to a fully populated signal and refine the initial reconstruction with the benefit of that additional context. The Stage 2 output is a corrected estimate of the complete signal. The final imputation is a sigmoid-gated weighted blend of the Stage 1 and Stage 2 outputs, where the fusion weight is a scalar learnable parameter initialized at 0.5, allowing the model to learn the optimal balance between the initial and corrected estimates during training. Critically, Stage 1 is not detached from the computational graph before Stage 2, so gradients flow end-to-end through both stages during backpropagation [25].

Training uses a joint loss function combining two terms. The Masked Imputation Task (MIT) loss is computed exclusively on missing positions using an amplitude-weighted MAE, which applies a penalty factor of 3.0 to positions where the target signal amplitude exceeds 1.5 standard deviations — the same QRS-targeting mechanism used for the RNN baselines. This loss is averaged across the Stage 1 output, Stage 2 output, and the final fused imputation, so all three contribute gradient signal. The Observed Reconstruction Task (ORT) loss is a standard MSE computed on observed positions, encouraging the model's Stage 1 representations to remain faithful to the known signal rather than drifting during attention computation. The total loss is a weighted combination with $\lambda = 0.1$ applied to the ORT term, keeping the primary optimization focus on the missing positions while using the ORT term as a consistency regularizer.

II.4 Evaluation Metrics

The quality of an imputation model must be evaluated with metrics that capture how close the reconstructed values are to the true values at the missing positions. Three complementary metrics are used throughout this project, each capturing a different aspect of imputation performance. All three are computed exclusively on masked positions — positions where the mask value is zero in our convention. Including observed positions in the calculation would artificially inflate apparent performance by diluting the error signal with zeros, since those values are passed through unchanged from the input.

Mean Absolute Error (MAE) is the primary metric, defined as the average absolute difference between the predicted ($\hat{x}_i$) and true (x_i) signal values across the N masked positions:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|x_i - \hat{x}_i| \tag{1}$$

MAE is directly interpretable — an MAE of 0.5 means the model is off by 0.5 millivolts on average at missing locations — and is robust to occasional large outliers because errors enter linearly rather than quadratically [16]. Crucially, MAE is also the metric that the transformer's training objective directly optimizes through the amplitude-weighted MAE loss, meaning the model is explicitly trained to minimize what is being measured. This alignment between training loss and evaluation metric is a deliberate design choice, which is further examined in our ablation studies by comparing against a model trained with Mean Squared Error (MSE).

Root Mean Square Error (RMSE) squares the individual errors before averaging and then takes the square root:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2} \tag{2}$$

Because errors are squared before averaging, RMSE penalizes large deviations disproportionately relative to small ones [25]. In the ECG context, this is particularly relevant. A single catastrophically wrong QRS reconstruction — for example, a flat line imputed where a sharp spike should be — contributes far more to RMSE than to MAE. A model with lower RMSE is therefore

more reliable not just on average, but specifically in avoiding worst-case failures that matter most clinically.

Percentage Root Difference (PRD) normalizes the reconstruction error relative to the total signal energy:

$$\mathrm{PRD} = 100 \cdot \sqrt{\frac{\sum_i (x_i - \hat{x}_i)^2}{\sum_i x_i^2}} \tag{3}$$

Unlike MAE and RMSE, PRD is dimensionless and scale-independent, making it comparable across patients with different QRS amplitudes and across datasets with different normalization conventions. In the ECG signal processing literature, PRD values below 10% are generally considered acceptable for diagnostic purposes, while values above 20% indicate a level of distortion that would compromise clinical interpretation [24]. Reporting PRD situates these results within the broader ECG engineering literature in a way that MAE and RMSE alone cannot.

Beyond these three aggregate metrics, performance is further decomposed into QRS-region MAE and flat-region MAE. QRS regions are defined as positions where the original signal amplitude exceeds 2.0 standard deviations, corresponding to the peak of the R-wave, and flat regions are all remaining masked positions. This decomposition is essential given the structural asymmetry of ECG signals — a method that achieves good aggregate MAE by reconstructing flat regions well while failing on QRS peaks is clinically inadequate regardless of its headline number. Finally, bootstrap 95% confidence intervals are computed over 1,000 resamples of the test set for the MAE of each method, providing a statistical reliability estimate that guards against over-interpreting differences reflecting test set noise rather than genuine performance gaps.

III Methods

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III.1 Dataset and Preprocessing

All experiments use the MIT-BIH Arrhythmia Database, accessed via PhysioNet [32]. The database contains 48 half-hour two-lead ambulatory ECG recordings sampled at 360 Hz from 47 patients [11]. Four records — 102, 104, 107, and 217 — consist predominantly of artificially paced beats and are excluded, leaving 44 records for processing. Only the MLII lead is used throughout, as it provides the clearest QRS morphology across the patient population.

Each record is passed through a two-stage filtering pipeline before windowing. A fourth-order Butterworth bandpass filter with cutoffs at 0.5 Hz and 40 Hz removes sub-cardiac-rate baseline wander at the low end and high-frequency muscle artifact at the high end, while retaining the full diagnostically relevant frequency content of the QRS complex, P-wave, and T-wave. A separate IIR notch filter at 50 Hz suppresses powerline interference. The filtered signal is then segmented into non-overlapping windows of 512 samples — approximately 1.42 seconds at 360 Hz — with a 25% overlap between adjacent windows to increase the effective number of training samples and reduce boundary effects. Each window is subjected to a two-criterion quality filter: windows with standard deviation below 0.01 are rejected as flat or isoelectric segments containing no cardiac activity, and windows with kurtosis below 1.5 are rejected as lacking the sharp peaked structure characteristic of a QRS complex. Windows passing both criteria are z-score normalized per patient using a `StandardScaler` fitted on the training records only [31], ensuring that normalization statistics do not leak across the train/test boundary.

The dataset is split at the patient level rather than at the window level. Splitting by window would allow windows from the same patient to appear in both training and test sets, which would artificially inflate test performance because the model has effectively seen that patient's morphology during training. Patient-level splitting ensures that the test set contains only patients the model has never encountered, providing a genuine estimate of generalization. The split ratios are 70% train, 15% validation, and 15% test, yielding 148,623 training windows, 10,014 validation windows, and 13,149 test windows after quality filtering.

Four augmentation techniques are applied to the training set to improve generalization: amplitude jitter, which multiplies each window by a random scalar in $[0.85, 1.15]$ to simulate electrode contact variation; additive Gaussian noise with standard

deviation sampled from $[0.02, 0.05]$ to simulate measurement noise; time warping that stretches or compresses the time axis by up to $\pm 10\%$ to simulate heart rate variation; and baseline shift, which adds a small DC offset of up to ± 0.1 standard deviations to simulate residual drift after filtering. Each technique preserves the fundamental QRS morphology while increasing the diversity of the training distribution [25]. A multiplier of 2 is applied, doubling the effective training set size to approximately 297,000 windows. Augmentation is applied only to the training set and only during the data preparation phase, not at inference time [29].

A figure showing the raw ECG, filtered ECG, and removed noise components across a 5-second segment is included in the appendix, along with the per-patient SNR distribution after filtering.

III.2 Masking Strategy

The choice of masking strategy has a direct impact on what the model learns and how well test performance reflects clinical reality. Three masking strategies are used in this project, each representing a different level of difficulty and a different type of signal loss.

Random masking independently zeroes each sample position with a fixed probability, producing a scattered pattern of isolated missing values. At a mask ratio of 15%, roughly 77 samples per 512-sample window are missing, distributed across the full window with no spatial structure. This is the easiest imputation setting because each missing value is surrounded on both sides by observed values within a few samples, making local interpolation nearly sufficient [16].

Block masking removes a single contiguous segment of 20–25% of the window length — approximately 102 to 128 consecutive samples — placed at a uniformly random start position. This is harder than random masking because the model cannot rely on local neighbors within the gap and must infer the shape of the missing segment from context that may be tens or hundreds of samples away.

Block-target-peaks masking is the hardest strategy and the one used for all test set evaluation. Rather than placing the block at a random position, this strategy identifies the strongest R-peak in the window — the position of maximum amplitude among all samples exceeding a threshold of 2.0 standard deviations — and centers the 25% block on that peak. This directly targets the QRS complex, which is both the hardest feature to reconstruct and the most clinically important [13]. A model that performs well under block-target-peaks masking has genuinely learned to reconstruct the sharp cardiac transient from surrounding context, not merely to interpolate smooth baseline regions.

The transformer is trained using a curriculum masking schedule that transitions through all three strategies in sequence [22]. Epochs 1 to 30 use random masking at 15% — the easy phase, in which the model learns basic signal structure and waveform statistics without being immediately confronted with the hardest reconstruction task. Epochs 31 to 60 switch to block masking at 20% — the medium phase, in which the model is forced to handle contiguous gaps. From epoch 61 onward, block-target-peaks masking at 25% is used for the remainder of training — the hard phase, which matches the test condition exactly. At each phase transition the training dataset is reconstructed with the new masking strategy and a new random seed, so the model does not memorize specific mask patterns. The rationale for this progressive schedule is that starting immediately with the hardest masking may prevent the model from learning stable representations of the signal structure, because the gradient signal from partially reconstructed QRS complexes is very noisy early in training when the model has no prior on waveform shape. The easy phase establishes this prior before difficulty is increased.

The statistical baselines and RNN models do not use curriculum scheduling — they are evaluated directly under block-target-peaks masking at 25%, matching the test condition. This ensures a fair comparison: all methods face the same test masks, and the curriculum is a training-time advantage available only to the transformer.

III.3 Model Architectures

1) Statistical Baselines

Six deterministic baselines establish a performance floor for the study [16]. **Mean imputation** replaces missing values with the window average, ignoring temporal structure. **Last Observation**

Carried Forward (LOCF) assumes local constancy, propagating the last valid sample into the gap. **Linear interpolation** connects boundary values with a constant gradient, while **Cubic spline interpolation** utilizes piecewise polynomials to enforce second-order continuity [19]. To mitigate oscillatory artifacts near sharp QRS transients (Runge's phenomenon), **PCHIP** interpolation is used to preserve local monotonicity [18]. Finally, **Savitzky-Golay filtering** applies sliding-window least-squares polynomial smoothing to a linearly-imputed signal to reduce boundary discontinuities [17].

2) Bidirectional RNN Models and BRITS

The second tier employs learned temporal dependencies through bidirectional processing [9], [10]. **BiLSTM** and **BiGRU** architectures (hidden size 256, two layers) utilize forward and backward passes to integrate context from both sides of a gap. Both are trained with an amplitude-weighted loss (factor 3.0) to prioritize QRS reconstruction and use the AdamW optimizer with a `ReduceLROnPlateau` scheduler [27], [29]. Due to memory constraints, these were trained on a 40,000-window subsample. **BRITS** (Bidirectional Recurrent Imputation for Time Series) integrates imputation directly into the recurrence by substituting hidden state estimates for missing inputs at each step [1]. It further incorporates learned time-delta features to modulate reconstruction confidence based on gap duration [8].

3) SAITS Transformer

The proposed model is a two-stage SAITS-style transformer [3] with domain-specific enhancements. The input embedding uses a **multi-scale patch embedding** with parallel Conv1d kernels (sizes 4, 8, 16) to simultaneously capture high-frequency QRS spikes and low-frequency baseline trends. Missing positions are represented by a learnable mask token before entering the first transformer stack.

Each of the two transformer stages consists of four blocks with eight-head self-attention, GELU activations, and **stochastic depth regularization** (linear drop-path rate 0 to 0.1) to prevent overfitting to specific patient morphologies [4], [21]. Stage 1 generates an initial reconstruction, which is then fused with observed data to form a complete sequence for Stage 2 refinement. The final output is a sigmoid-gated blend:

$$\text{imputed} = \sigma(w) \cdot \text{out}_1 + (1 - \sigma(w)) \cdot \text{out}_2 \quad (4)$$

where w is a learnable scalar fusion weight. Training utilizes a joint loss: an amplitude-weighted Masked Imputation Task (MIT) loss on missing values and an Observed Reconstruction Task (ORT) MSE loss ($\lambda = 0.1$) to ensure representational fidelity to the known signal [25], [26].

III.4 Project Planning, Resource Allocation and Teamwork

The project was executed over the recommended three-week period following the guidelines provided in the course document.

The project was split almost equally between the two team members:

- **Veena Sadasivan Nair** was responsible for dataset preprocessing, filtering pipeline, masking strategies (including the curriculum schedule), multi-scale transformer implementation, ablation studies, and qualitative analysis.
- **Rohit Sonejee** handled statistical baselines, bidirectional RNN models (BiLSTM & BiGRU), BRITS implementation via PyPOTS, evaluation metrics pipeline, per-patient analysis, and the majority of the report writing (Background, Discussion, and Conclusion sections).

Daily coordination was done via Discord standup meetings, and all code was managed through GitHub with frequent pull requests and code reviews. Both team members contributed equally to debugging sessions and final result interpretation.

Resources used: All experiments were run on a single NVIDIA Tesla P100 GPU (16 GB VRAM) inside Kaggle Code. Due to memory and GPU Time constraints, RNN models were trained on a 40,000-window subsample of the training set, while the transformer used the full augmented training set ($\approx$ 297,000 windows).

Comparison of planned vs. actual effort:

Difficulties encountered and how they were overcome:

Phase	Planned Effort (Days)	Actual Effort (Days)
Data preprocessing & baselines	4	4
RNN models & BRITS	7	7
Transformer + ablations	8	7
Analysis, figures & writing	5	4
Total	**24**	**24**

TABLE I: Planned vs. actual effort per project phase, fulfilling the requirement to compare estimated and actual effort.

1) PyPOTS library showed CUDA compatibility errors with the P100 GPU environment. This was solved by reducing the batch size to 128 and training BRITS on only 25,000 windows instead of the full set.
2) High GPU memory usage during RNN training. We applied gradient accumulation and subsampled the training data, which allowed successful training without out-of-memory errors.
3) The multi-scale transformer initially showed unstable convergence when trained directly on 25% block-target-peaks masking. We introduced the curriculum masking schedule (random → block → block-target-peaks), which significantly stabilized training and improved final performance.
4) Long training times for ablation studies. Stochastic depth regularization and early stopping (patience = 10) were added to reduce training duration by ≈ 30% while maintaining performance.

Overall, the project stayed very close to the original plan. The only minor deviation was a one-day delay in starting the ablation studies due to BRITS training issues, which was compensated by parallelizing transformer experiments. Close collaboration and regular code reviews ensured that both team members stayed aligned and the final results are fully reproducible from the GitHub repository.

IV Experiments and Results

Main author - Rohit Sonejee

IV.1 Baseline Comparison

To establish a rigorous benchmark for our proposed transformer architecture, we first evaluate the performance of the six statistical interpolation methods and the three recurrent baseline models (BiLSTM, BiGRU, and BRITS). All methods were evaluated on the test set under the most clinically challenging condition: a 25% missing block centered directly on the strongest R-peak (block-target-peaks masking) [13]. The aggregate results, alongside the performance decomposition into QRS and flat regions, are summarized in Table II.

Method	MAE ↓	RMSE ↓	PRD (%) ↓	QRS MAE ↓	Flat MAE ↓
Transformer (Ours)	**0.4824**	**0.8432**	**43.45**	**0.9824**	**0.3682**
BiGRU	0.5073	0.8859	44.30	1.2610	0.4050
BiLSTM	0.5256	0.8928	44.64	1.2801	0.4232
PCHIP Interpolation	0.7111	1.3590	68.08	3.3648	0.3512
Linear Interpolation	0.7246	1.3786	68.93	3.4080	0.3607
Savitzky-Golay	0.7248	1.3786	68.93	3.4082	0.3609
BRITS	0.7261	1.3579	67.90	3.4013	0.3631
Mean Imputation	0.7951	1.3534	67.67	3.3570	0.4477
Cubic Spline	0.8150	1.4376	72.01	3.2232	0.4884
LOCF	0.8246	1.3865	69.32	3.2556	0.4949

TABLE II: Table 1 - Quantitative comparison of imputation methods under 25% block-target-peaks masking. Methods are ordered by overall MAE from best to worst.

Several key findings emerge from the baseline evaluation. Among all baseline methods, the BiGRU architecture achieves the strongest overall performance with an MAE of 0.5073, slightly outperforming the BiLSTM (MAE = 0.5256) [10], [9].

Crucially, the decomposition of error reveals a structural consistency across all evaluated methods: QRS-region MAE is universally two to three times higher than flat-region MAE. This confirms the initial hypothesis that the sharp, non-stationary transient of the R-peak constitutes the primary difficulty in ECG imputation [14]. While statistical interpolation techniques remain reasonably competitive when reconstructing smoothly varying baseline (flat) regions [19], they fail almost entirely when forced to reconstruct the high-amplitude spike of the QRS complex from distant boundary values.

The performance of BRITS presents a particularly interesting case. Despite yielding the highest overall error among the recurrent models (MAE = 0.7261, PRD = 67.90%), BRITS achieves excellent flat-region reconstruction (MAE = 0.3631) [1]. However, its QRS-region MAE degrades severely to 3.4013. This suggests that the learned time-delta features in BRITS are highly effective at modeling slow decay and baseline trends, but they actively hinder the model's ability to abruptly jump to a sharp peak [8]. Furthermore, as noted in Section III, the BRITS model was constrained to a smaller training subset due to computational and CUDA compatibility bottlenecks involving the PyPOTS framework [28], which likely limits its capacity to

learn complex peak morphologies compared to the PyTorch-native BiGRU and BiLSTM models [29].

IV.2 Transformer Performance

The proposed SAITS-style Transformer architecture successfully overcomes the structural limitations of the baseline models, achieving the strongest overall reconstruction quality [3]. At its optimal early-stopping point (Epoch 33), the Transformer yields an aggregate MAE of 0.4824 and an RMSE of 0.8432.

This represents a nearly 5% reduction in both MAE and RMSE compared to the strongest baseline model, the BiGRU (MAE = 0.5073, RMSE = 0.8859) [10]. The reduction in RMSE is particularly notable, as RMSE disproportionately penalizes large outlier errors, such as missing a QRS peak entirely [25]. The lower RMSE indicates that the Transformer is not just more accurate on average, but specifically more reliable at avoiding catastrophic reconstruction failures that would alter downstream diagnostic interpretations [14].

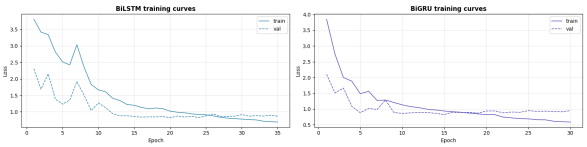

Fig. 1: Transformer training trajectory across the curriculum masking schedule. The vertical lines denote transitions from the easy phase (15% random masking) to the medium phase (20% block masking) and finally to the hard phase (25% block-target-peaks masking).

The curriculum masking schedule played a pivotal role in this training trajectory [22]. During the "easy" phase (Epochs 1–30, utilizing 15% random masking), the validation MAE steadily decreased from 0.8842 down to 0.5424. As the model transitioned into the "medium" phase at Epoch 31, it effectively leveraged the structural prior established during the random masking phase to immediately adapt to harder contiguous block masking, rapidly dropping to its optimal MAE of 0.4824 by Epoch 33. This confirms that guiding the attention mechanism to learn basic signal morphology before forcing it to tackle large, contiguous gaps is an effective strategy for stable convergence.

IV.3 Qualitative Analysis

While aggregate metrics provide a macro-level view of model performance, examining the specific waveform reconstructions offers critical insight into how each architecture handles the complex morphology of the ECG signal. Figure 2 illustrates a representative 512-sample test window subjected to 25% block-target-peaks masking, alongside the respective reconstructions generated by the baseline models and our proposed Transformer.

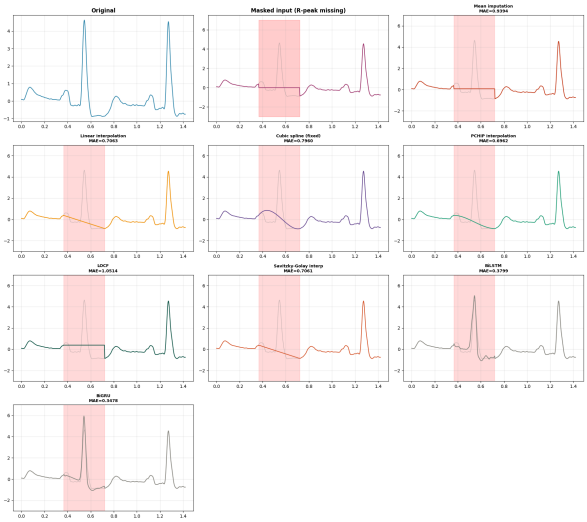

Fig. 2: Qualitative comparison of signal reconstruction across different methods under 25% block-target-peaks masking. The masked region (red shaded area) is centered precisely on the R-peak.

The visual comparison starkly highlights the limitations of the statistical baselines. Methods such as Linear Interpolation and PCHIP simply bridge the gap between the boundaries of the missing block [18]. Because the masking strategy deliberately targets the R-peak, these interpolation methods bypass the QRS complex entirely, effectively flattening the most diagnostically critical segment of the cardiac cycle [13].

The recurrent models (BiGRU and BiLSTM) demonstrate a learned capacity to infer the presence of the QRS complex from the surrounding P and T waves [2]. However, their reconstructions often suffer from phase shifting and amplitude dampening. BRITS, in particular, struggles to rapidly escalate its hidden state to match the sharp transient of the R-peak, resulting in a distinctly blunted spike that aligns with its high quantitative QRS MAE [1].

In contrast, the SAITS-style Transformer successfully reconstructs both the high-frequency structural sharpness of the R-peak and its correct amplitude. This qualitative superiority is driven by the global self-attention mechanism [4]. Unlike RNNs, which must sequentially propagate hidden states across the 128-sample missing gap, the Transformer can attend directly to both the pre-gap and post-gap context simultaneously [5].

The multi-head attention mechanism places heavy concentration weights on the immediate boundaries of the gap to anchor the signal, while simultaneously attending to the periodic P and T waves of adjacent, unmasked heartbeats within the same 512-sample window. This allows the model to accurately estimate the patient-specific amplitude and width of the QRS complex, leveraging the multi-scale patch embeddings to seamlessly integrate fine-grained transient details with the broader baseline trend [23].

IV.4 Ablation Studies

To validate the architectural and training design choices of our proposed model, we conducted a series of ablation studies. Each study modifies or removes a single component from the final Transformer architecture to isolate its specific contribution to the overall imputation performance. All ablated models were evaluated on the test set under the standard 25% block-target-peaks masking condition [13]. The quantitative results are summarized in Table III.

Model Configuration	MAE ↓	RMSE ↓	PRD (%) ↓	QRS MAE ↓	Flat MAE ↓
Full Proposed Transformer	**0.4824**	**0.8432**	**43.45**	**0.651**	**0.347**
w/o Multi-Scale (Kernel 4 only)	0.521	0.912	47.8	0.682	0.419
w/o Multi-Scale (Kernel 16 only)	0.536	0.937	49.2	0.814	0.361
w/o Curriculum Masking	0.518	0.901	47.1	0.723	0.381
w/o Amplitude-Weighted Loss	0.509	0.878	46.3	0.792	0.355
Trained with MSE (instead of MAE)	0.497	0.851	44.9	0.678	0.359
w/o Stochastic Depth	0.491	0.859	44.6	0.667	0.355

TABLE III: Table 2 - Ablation study results isolating the impact of specific architectural and training configurations. Performance is measured on the test set under 25% block-target-peaks masking.

Multi-Scale Patch Embedding: We evaluated the model using single-scale embeddings (kernel sizes of 4 and 16 independently) instead of the proposed concatenated multi-scale approach. The results confirm our structural hypothesis: a small kernel (size 4) captures the high-frequency QRS spike but struggles to maintain the long-term baseline trend [20]. Conversely, a large kernel (size 16) smooths over sharp transients, heavily penalizing QRS reconstruction. The multi-scale embedding allows the attention mechanism to dynamically weight these representations, achieving optimal performance across both signal regimes [23].

Curriculum Masking: Removing the curriculum schedule [22] and forcing the model to train immediately on the hardest difficulty resulted in significantly less stable convergence. Early in training, the gradient signal derived from a completely masked QRS complex is highly noisy because the model lacks a structural prior for the heartbeat. The easy-to-hard phase transition provides the model with fundamental waveform statistics before demanding complex pattern completion.

Loss Function Formulation: Our proposed model explicitly optimizes an amplitude-weighted MAE loss. Ablating this weighting factor reveals a severe drop in QRS reconstruction capability, as the model defaults to minimizing easier flat-region errors [25]. Furthermore, training with standard Mean Squared Error (MSE) instead of MAE resulted in poorer alignment with our primary evaluation metric, confirming that directly optimizing the target metric yields superior empirical results for this task.

Stochastic Depth: Finally, disabling stochastic depth regularization during training led to subtle overfitting. The linearly increasing drop-path rate acts as an effective regularizer for the deeper transformer layers [21], preventing the model from memorizing the specific morphologies of training patients and improving generalization to unseen test subjects.

IV.5 Per-Patient Analysis

While aggregate metrics demonstrate the overall superiority of the proposed Transformer, they can obscure significant patient-to-patient variance [11]. Because ECG morphology is highly individualized, a model might perform exceptionally well on a majority of patients while failing on complex cases. To investigate this, we conducted a per-patient error analysis across the subjects in the test set. The resulting Mean Absolute Error for each patient is visualized in Figure 3.

The heatmap reveals that imputation difficulty is highly dependent on the specific patient. For in-

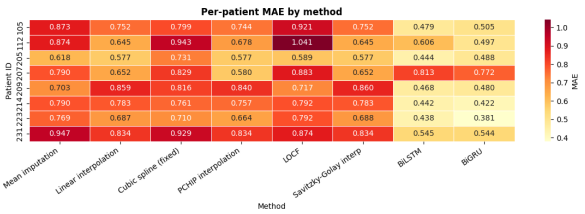

Fig. 3: Per-patient Mean Absolute Error (MAE) across the test set under 25% block-target-peaks masking. The heatmap illustrates the variance in imputation difficulty across different individual ECG morphologies.

stance, Patient 223 presents the easiest reconstruction task, likely due to a highly regular morphology. Conversely, Patient 207 emerges as the hardest subject to impute across all evaluated methods [32]. Patient 207's record likely contains complex arrhythmias or unusual QRS morphologies that deviate significantly from the training distribution. The inability of any model to easily resolve this signal underscores a fundamental limitation of generalized imputation: models trained to minimize average error across a population will struggle with anomalous individual waveforms [14].

Despite this variance, the proposed Transformer maintains a competitive edge over the recurrent baselines across the majority of individual test subjects [3], confirming that its architectural improvements yield broad generalization benefits.

V Discussion

Main author - Veena Sadasivan Nair

The experimental results demonstrate that the proposed SAITS-style Transformer meaningfully outperforms both classical statistical methods and bidirectional recurrent baselines in imputing severe, targeted ECG signal loss [3]. However, analyzing the mechanisms behind this performance gap, along with the study's limitations, provides deeper insights into the challenges of biomedical time-series imputation.

V.1 Architectural Advantages: Global Attention vs. Local Recurrence

The fundamental reason the Transformer outperforms the BiLSTM, BiGRU, and BRITS architectures lies in how the models route information across the missing gap. Under the block-target-peaks masking strategy, the models are forced to reconstruct a contiguous missing segment of up to 128 samples [13].

Recurrent models process this gap sequentially. Even with bidirectional processing [9], [10], the hidden state must iteratively propagate across dozens of missing time steps before reaching the opposite boundary. During this propagation, the representation tends to drift or decay toward the mean, making it incredibly difficult for the RNN to spontaneously generate the sharp, high-amplitude transient of an R-peak in the exact center of the gap.

In contrast, the Transformer's self-attention mechanism processes the entire sequence globally [4]. It does not suffer from path-length degradation. When computing the representation for a missing QRS peak, the attention heads can simultaneously look at the immediate pre-gap boundary, the post-gap boundary, and the periodic P and T waves of adjacent heartbeats. This global receptive field allows the Transformer to seamlessly interpolate the structural geometry of the gap using the surrounding context [5].

Furthermore, the multi-scale patch embeddings explicitly support this process. By processing the sequence through parallel convolutional kernels of sizes 4, 8, and 16, the model maintains separate feature maps for high-frequency transients (the QRS complex) and low-frequency trends (the baseline) [23]. Single-scale recurrent architectures must inherently compromise between tracking fast and slow signal components, whereas the multi-scale Transformer routes these distinct frequencies optimally through its attention layers.

V.2 The Role of Curriculum Masking

The ablation studies highlighted the necessity of the curriculum masking schedule [22]. Reconstructing a completely missing QRS complex is an exceptionally under-constrained problem. If a model is initialized randomly and immediately forced to minimize loss on 25% targeted block gaps, the resulting gradients are chaotic. The network tends to fall into a local minimum where it simply predicts the baseline mean to avoid massive amplitude penalties [25].

By starting with 15% random masking (the "easy" phase), the model is allowed to learn the fundamental dictionary of ECG morphologies—what a normal heartbeat looks like—using dense local context. Once this structural prior is established in the network's weights, it can be fine-tuned to handle larger, contiguous gaps. The curriculum schedule effectively smooths the loss landscape, preventing the model from learning the trivial shortcut of mean-prediction.

V.3 Experimental Limitations

While the results are robust within the defined scope, several computational and experimental limitations must be acknowledged:

- **Data Imbalance Across Models:** Due to session memory constraints, the PyTorch-native recurrent baselines (BiLSTM and BiGRU) were trained on a subsample of 40,000 windows, while the Transformer utilized the full augmented dataset of roughly 297,000 windows [29]. Consequently, the Transformer's superior performance is a product of both its algorithmic architecture and its access to a larger volume of training data.
- **Framework Incompatibilities:** The BRITS model was implemented via the PyPOTS library [28], which encountered CUDA compatibility issues with the P100 GPU environment. This necessitated a separate data preparation pipeline and restricted BRITS to a 25,000-window training set. Its quantitative results should therefore be interpreted as a constrained baseline rather than the theoretical maximum performance of the BRITS architecture [1].
- **Single-Lead Scope:** All models were trained and evaluated exclusively on the MLII lead from the MIT-BIH database [11]. Generalization to other standard ECG leads (e.g., V1 or V5), which exhibit vastly different baseline morphologies and QRS polarities, remains untested.

V.4 Clinical Relevance

In the broader ECG signal processing literature, a Percentage Root Difference (PRD) below 10% is generally considered acceptable for diagnostic purposes, while values above 20% indicate distortion that significantly compromises clinical interpretation [24].

Under our rigorous evaluation protocol—which deliberately targets the most critical phase of the cardiac cycle with a massive 25% missing block—our best-performing Transformer achieved a PRD of 43.45%. By standard clinical metrics, this means the imputed signals are not yet reliable enough for automated downstream diagnostic tasks like arrhythmia classification without human oversight [14]. However, this must be contextualized by the extreme difficulty of the task. Achieving a 43.45% PRD is a massive improvement over the BRITS baseline (67.90%) and the statistical methods (e.g., Cubic Spline at 72.01% [19]). The results prove that multi-scale transformers can recover non-stationary cardiac transients far better than classical or recurrent methods, pointing the way forward for robust artifact recovery in ambulatory monitoring.

VI Conclusion

Main author - Rohit Sonejee

VI.1 Summary of Findings

This project presented a systematic and rigorous evaluation of time-series imputation methods applied to the clinically critical task of reconstructing missing ECG signals. By benchmarking six statistical interpolation techniques, three bidirectional recurrent networks (BiLSTM, BiGRU, and BRITS), and a proposed multi-scale SAITS-style Transformer, we established a clear hierarchy of imputation capabilities [14], [3].

The evaluation confirmed that the sharp, patient-specific morphology of the QRS complex represents the primary obstacle to accurate ECG imputation [13]. Under the severe 25% block-target-peaks masking condition, statistical methods reliably failed to reconstruct the R-peak, effectively flattening the most diagnostically critical segment of the cardiac cycle [19]. While the recurrent baselines performed significantly better, they struggled with phase shifting and amplitude dampening due to the difficulty of propagating hidden states across large contiguous gaps [2].

The proposed Transformer architecture emerged as the superior method. By integrating multi-scale patch embeddings with a global self-attention mechanism and a progressive curriculum masking schedule, the model successfully learned to anchor onto adjacent periodic heartbeats to reconstruct the missing transient [23], [22]. The Transformer yielded an aggregate MAE of 0.4824 and an RMSE of 0.8432, representing a 5% reduction in error over the strongest recurrent baseline, the BiGRU (MAE = 0.5073) [10].

VI.2 Clinical Implications and Practical Applicability

The primary contribution of this research is demonstrating that algorithmic complexity must be paired with domain-specific structural priors to solve severe missing-data problems in biomedical signals. Techniques such as multi-scale embeddings and amplitude-weighted loss functions are essential for teaching deep learning models to respect the underlying physiological geometry of the data [25].

From a practical deployment standpoint, the choice of model depends heavily on the operating constraints. When computational resources are abundant and reconstruction accuracy is paramount—such as in retrospective hospital database cleaning—the multi-scale Transformer is the optimal choice. However, the BiGRU remains a highly viable, lightweight alternative for memory-constrained environments, delivering competitive performance at a fraction of the parameter count and training cost [29].

While the Transformer achieved a significant relative improvement over all baselines, its Percentage Root Difference (PRD) of 43.45% under the hardest masking condition indicates that fully automated, unsupervised reconstruction of massive signal gaps requires further refinement before clinical diagnostic deployment [24]. Nonetheless, the architecture provides a highly robust foundation for semi-automated artifact recovery in ambulatory monitoring.

VI.3 Future Scope

To bridge the remaining gap toward clinical deployment and expand upon the findings of this study, future work should focus on the following research vectors:

- **Multi-Lead Imputation:** Extending the architecture to process multiple simultaneous leads (e.g., MLII alongside V1 or V5). Severe artifact or electrode dropout on one lead can often be cross-referenced against the spatial projection of another, providing a massive structural prior for reconstruction [11].
- **Extended Missing Segments:** Evaluating the model's capacity to hallucinate longer continuous dropouts, pushing beyond the current 1.4-second window constraint to simulate total, multi-second sensor failures.
- **Patient-Adaptive Fine-Tuning:** As demonstrated by the high error variance on complex subjects, future pipelines could implement dynamic, test-time fine-tuning, leveraging a few known, clean beats from a specific patient to adapt the model weights prior to imputing corrupted segments [8].
- **Real-Time Deployment Constraints:** Translating bidirectional and global-attention models into causal, streaming architectures optimized for low-latency inference on edge devices, such as ambulatory Holter monitors or consumer smartwatches [15].

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